Project Dashboard - Comprehensive Project Tracking System

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1. Project Basic Information

Project Name:	Plant Monitoring Sensor Network		
Client Name:	John bambery		
Start Date:	25-Oct-25	**Expected End Date:**	30-Nov-25
Pricing Model:	**Fixed Price** ☐ **Hourly Rate** ☑		
Hourly Rate Per Hour ($):	$20.87	**Total Project Budget:**	5000$
Payment by Milestone:	☐		

2. Project Requirements

Project Requirements:

The project involves designing and building an MVP prototype of a multi-modal plant monitoring system. It includes two battery-powered, weatherproof sensor nodes integrating environmental sensors (soil moisture, temperature, humidity, light, pH, atmospheric pressure) and plant communication sensors (bioelectric potential, VOCs, IR leaf temperature, optional ultrasonic). Each node must support data collection, logging, and LoRa transmission, with firmware for sensor calibration, power management, and basic visualization.

3. Project Milestones

	Start Date:			
	Task Requirements	**Hours Spent**	**Description**	**Status**
Milestone 1	Finalize System Architecture	12	Planning & Initial Testing	Complete
	Finalising Components and Create Shopping List	26		Complete
	Breadboard testing with Sensors (solderless)			
	Firmware Development with sensor interation			
	Real-World Unit Testing (Minimal Functionalities ofline Version)			
	Integrate battery & solar backup			
	Milestone 1 Hours :	38		
	Task Requirements	**Hours Spent**	**Description**	**Status**
Milestone 2	Add LoRa & IoT functionality		Power & Connectivity (IoT or LORA)	
	Develop central node / desktop backend for data preprocessing			
	Assemble waterproof enclosure & place sensors			
	Firmware enhancement (calibration, buffering, power management)			
	PCB assembly & solder components (Adding Indicators on Enclosure)			
	Milestone 2 Hours :	0		
	Task Requirements	**Hours Spent**	**Description**	**Status**
Milestone 3	System integration (hardware + firmware + sensors)		Final Assembly & Delivery	
	End-to-end testing (sensors, battery, LoRa, visualization)			
	Complete documentation (manuals, diagrams, instructions)			
	Complete manuals, diagrams, and assembly instructions.			
	Deliver final project to client			
	Milestone 3 Hours :	0		

4. Key Deliverables

		Additional Requirements	Project End Goal and Approval Criteria
Sensor Selection	☑	**Hardware**	All hardware components functioning to specification
Firmware Development	☑	LoRa communication or WIFI	Software meets All requirements with no critical bugs
Schematic Drawing	☐	Assembled sensor nodes	Client signs off on final deliverables
PCB Layout	☐	Power requirements	
Gerber Files	☐	Solar battery system	
Bill of Materials (BOM)	☐	**Software**	**Design**
PCB Fabrication Ordering	☐	Firmware sensor integration	IP67 weatherproof enclosure
PCB Assembly (soldering components)	☑	Sensor calibration option	Outdoor mounting system and Compact modular design
Board Testing	☐	LED status indicators	Corrosion-resistant materials
3D Enclosure Design	☐		

5. Unallocated Task Time Tracking

Task Description	Hours Spent
Hours:	**0**

6. Additional Expenses :

Item / Description	Category	Notes	Cost per Unit ($)	Units	Total ($)
PCB Fabrication (JLCPCB)	Fabrication				$0.00
Customs VAT	Customs/Tax				$0.00
Enclosure Printing	Outsourcing				$0.00
PCB Assembly	Assembly Service				$0.00
Components	Components LCSC				$0.00
Unit Testing Labor	Testing				$0.00
Delivery to Client	Logistics				$0.00
Others					$0.00
				Total Cost:	**$0.00**

7. Payment Section

Hourly Payments

Task / Milestone	Rate ($/hr)	Hours Worked	Amount ($)	Paid ($)	Remaining ($)
Milestone 1	$20.87	38	$793.06	$500.00	$293.06
Milestone 2	$20.87	0	$0.00		$0.00
Milestone 3	$20.87	0	$0.00		$0.00
Extra Tasks (Unallocated)	$20.87	0	$0.00		$0.00
Additional Expenses	Nil	Nil	$0.00		$0.00
Total Amount:		**38**	**$793.06**	**$500.00**	**$293.06**

Thank you for choosing to work with me. I truly appreciate your trust, and I'll do my best to bring your idea to life with precision, dedication, and transparency. This spreadsheet has been created to provide a clear overview of project scope, progress, and costs, ensuring full visibility at every step.

Please Note:

The quotation provided does not include external expenses such as:
- Component sourcing (AliExpress, Digi-Key, etc.)
- PCB fabrication or assembly services (e.g., JLCPCB, PCBWay)
- Import duties, VAT, or customs clearance fees
- Packaging or shipping of prototypes
- Specialized tooling or any outsourced service costs
- These costs will be discussed and billed separately if required during the project.

Looking forward to building something great together from concept to completion, I'm here as your technical partner.

Warm regards,
M. Zaeem
Electronics & Embedded Systems Engineer